RATING METHODOLOGY

15 November 2023

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Rating Methodology

Telecommunications Service Providers

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This methodology is no longer in effect. For more information on rating methodologies used by Moody's Investor Services, visit https://ratings.moodys.com/rating-methodologies.

This rating methodology replaces the *Telecommunications Service Providers* methodology published in September 2022 and the *Pay TV* methodology published in October 2021, which has been retired. While this methodology reflects many of the same core principles as the 2022 *Telecommunications Service Providers* methodology, we have expanded the methodology scope to include companies that were previously rated using the *Pay TV* methodology, because the end-market drivers and business profiles of companies in these sectors have converged. We have also made some changes to the scorecard. We have reduced the weight of the Scale and Business Profile factors and increased the weight of the Leverage and Coverage factor. Within the Business Profile factor, we have combined Business Model, Competitive Environment and Technical Positioning and Regulatory Environment into a single sub-factor named Competitive Position. Within the Leverage and Coverage factor, we have changed the RCF/Debt sub-factor to RCF/Net Debt. We have also changed some sub-factor descriptions, weights and thresholds and changed the presentation of the scorecard.

Scope

This methodology applies to companies globally that are primarily engaged in providing telecommunications services including data, voice and video services over wireless or wireline infrastructure, to other businesses or consumers.

The determination of a company's primary business is generally based on the preponderance of the company's business risks, which are usually proportionate to the company's revenues, earnings and cash flows.

This report was republished on October 4, 2024. We made a formatting change to incorporate a footnote in the "Scope" section into the body of the text. We also updated the exhibit sources.

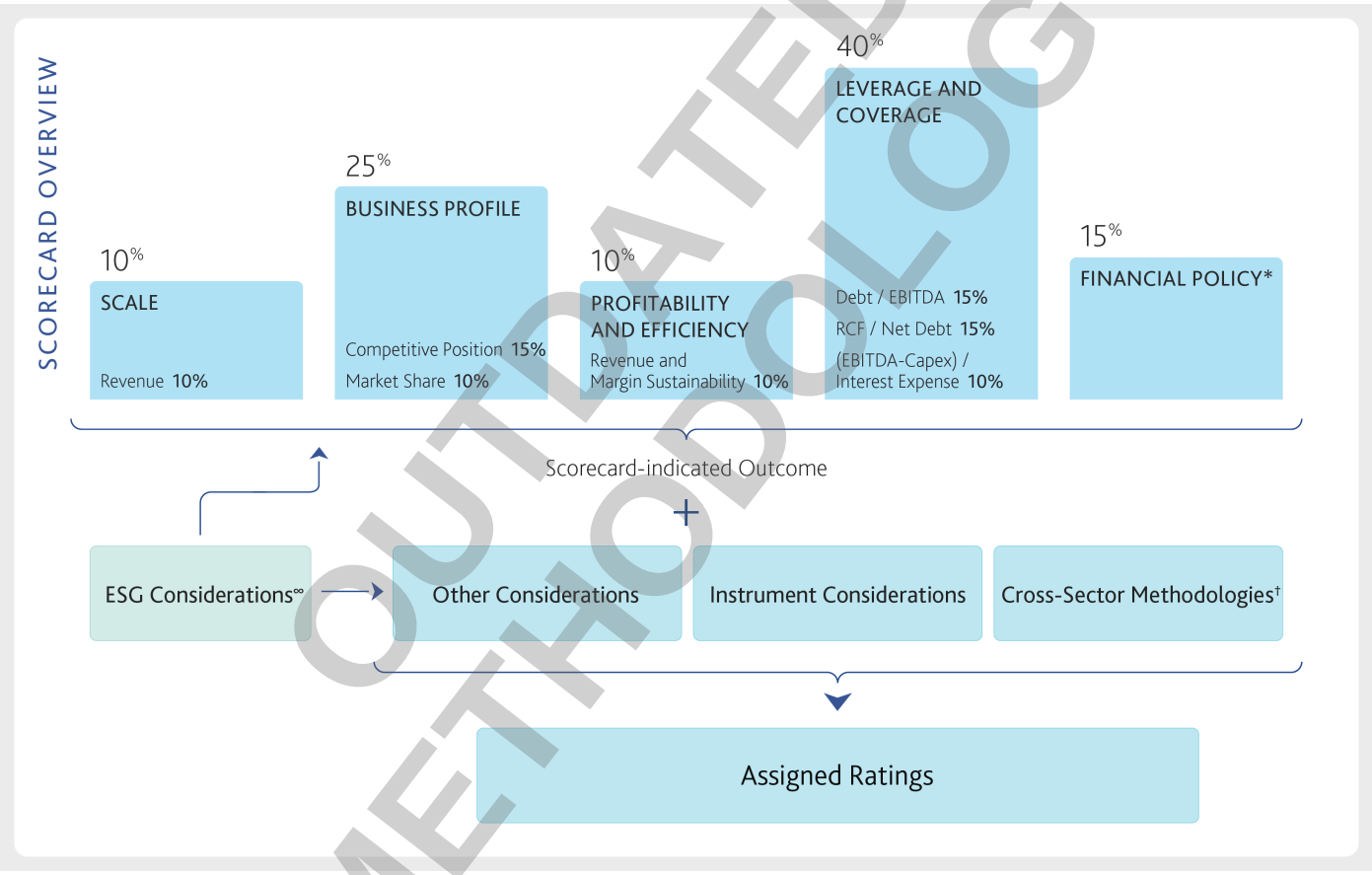
Rating approach

In this rating methodology, we explain our general approach to assessing credit risk of issuers in the telecommunications service provider industry globally, including the qualitative and quantitative factors that are likely to affect rating outcomes in this sector. We seek to incorporate all material credit considerations in ratings and to take the most forward-looking perspective that visibility into these risks and mitigants permits.

The following schematic illustrates our general framework for the analysis of telecommunications service providers, which includes the use of a scorecard. The scorecard-indicated outcome is not expected to match the actual rating for each company. For more information, see the "Other considerations" and "Limitations" sections.

Exhibit 1

Illustration of the telecommunications service providers methodology framework



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* This factor has no sub-factors.

∞ Environmental, social and governance (ESG) considerations, including, where available, our opinions of exposure to them as expressed in Issuer Profile Scores (IPSs), may affect scorecard factors and other considerations outside of the scorecard. For more information, see the "Other considerations" section.

† Some of the methodological considerations described in one or more cross-sector rating methodologies may be relevant to ratings in this sector. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

Source: Moody's Ratings

Telecommunications service providers scorecard

For general information about how we use the scorecard and for a discussion of scorecard mechanics, please see the "Using the scorecard to arrive at a scorecard-indicated outcome" section. The scorecard does not include or address every factor that a rating committee may consider in assigning ratings in this sector. Please see the "Other considerations" and "Limitations" sections.

Exhibit 2

Telecommunications service providers scorecard

	Weight	Aaa	Aa	A	Baa	Ba	B	Caa	Ca
Factor: Scale (10%)									
Revenue ($ billion)[1]	10%	≥ $100	$50 - $100	$25 - $50	$10 - $25	$4 - $10	$1.5 - $4	$0.5 - $1.5	< $0.5
Factor: Business Profile (25%)									
Competitive Position	15%	Provider of a full suite of telecommunications services; has an extremely broad customer base and extremely high geographic diversity; is exposed to effectively no competition; operates in an extremely supportive and predictable regulatory environment; all of its infrastructure and technology are industry-leading.	Provider of a full suite of telecommunications services; has a very broad customer base and very broad geographic diversity; is exposed to very limited competition; operates in a very supportive and predictable regulatory environment; almost all of its infrastructure and technology are industry-leading.	Provider of a broad suite of telecommunications services; has a broad customer base and broad geographic diversity; is exposed to limited competition; operates in a moderately supportive and predictable regulatory environment; most of its infrastructure and technology is industry-leading.	Provider of a broad suite of telecommunications services; has a significant customer base and good geographic diversity; is exposed to moderate competition; operates in a neutral regulatory environment; some of its infrastructure and technology is older generation.	Provider of a moderate suite of telecommunications services; has a moderate-sized customer base and moderate geographic diversity; is exposed to significant competition; operates in a moderately unsupportive regulatory environment; a significant portion of its infrastructure and technology is older generation.	Provider of a limited suite of telecommunications services; has a small customer base and limited geographic diversity; is exposed to significant and increasing competition; operates in a considerably unsupportive regulatory environment; most of its infrastructure and technology is older generation.	Provider of a very limited number of telecommunications services; has a very small customer base and very limited geographic diversity; is exposed to high and increasing competition; operates in a very unsupportive regulatory environment; all of its infrastructure and technology is older generation.	Provider of a single service; has an extremely small customer base and essentially no geographic diversity; is exposed to very high and increasing competition; operates in an extremely unsupportive regulatory environment; high probability of disruption in service because of poor infrastructure and technology.

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	Weight	Aaa	Aa	A	Baa	Ba	B	Caa	Ca
Factor: Business Profile (25%)									
Market Share	10%	The principal player in its local market and in most of the regions where it operates; or has monopoly-type presence in its local region.	A clear market leader in the local market and holds competitive positions in all regions where it operates; or the principal player and very strong market leader in its local region.	A very solid competitor in the local market and holds competitive positions in most of the regions where it operates; OR a clear market leader in its local region.	A well-positioned competitor in its local market and holds competitive positions in many regions where it operates; OR a very solid competitor in its local region.	A mid to lower-tier competitor in its local market and holds competitive positions in some of the markets where it operates; OR a well-positioned competitor in its local region.	A small competitor in its local market and holds minor competitive positions in other markets; OR a mid to lower-tier competitor in its local region.	A small competitor in its local market.	A start-up with no track record.
Factor: Profitability and Efficiency (10%)									
Revenue and Margin Sustainability	10%	Very strong, sustainable revenue growth; Extremely strong EBITDA margins.	Strong, sustainable revenue growth; Very strong EBITDA margins.	Moderate, sustainable revenue growth; Strong EBITDA margins.	Slight, sustainable revenue growth; Solid EBITDA margins.	Expectation of flat revenue; Moderate EBITDA margins.	Expectation of slight revenue decline; weak EBITDA margins.	Expectation of moderate revenue decline; very weak EBITDA margins.	Expectation of significant revenue decline; extremely weak EBITDA margins.
Factor: Leverage and Coverage (40%)									
Debt / EBITDA[2]	15%	≤ 0.5x	0.5x - 1x	1x - 2x	2x - 3x	3x - 4x	4x - 6x	6x - 8x	> 8x
RCF / Net Debt[3]	15%	≥ 90%	75% - 90%	55% - 75%	35% - 55%	20% - 35%	10% - 20%	5% - 10%	< 5%
(EBITDA-Capex) / Interest Expense[4]	10%	≥ 8x	6.5x - 8x	5x - 6.5x	3.5x - 5x	2x - 3.5x	1x - 2x	0.5x - 1x	< 0.5x

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	Weight	Aaa	Aa	A	Baa	Ba	B	Caa	Ca
Factor: Financial Policy (15%)									
Financial Policy	15%	Expected to have extremely conservative financial policies (including risk and liquidity management); very stable metrics; essentially no event risk that would cause a rating transition; and public commitment to a very strong credit profile over the long term.	Expected to have very conservative financial policies (including risk and liquidity management); stable metrics; minimal event risk that would cause a rating transition; and public commitment to a strong credit profile over the long term.	Expected to have predictable financial policies (including risk and liquidity management) that preserve creditor interests; although modest event risk exists, the effect on leverage is likely to be small and temporary; strong commitment to a solid credit profile.	Expected to have financial policies (including risk and liquidity management) that balance the interests of creditors and shareholders; some risk that debt-funded acquisitions or shareholder distributions could lead to a weaker credit profile.	Expected to have financial policies (including risk and liquidity management) that tend to favor shareholders over creditors; above-average financial risk resulting from shareholder distributions, acquisitions, or other significant capital structure changes.	Expected to have financial policies (including risk and liquidity management) that favor shareholders over creditors; high financial risk resulting from shareholder distributions, acquisitions, or other significant capital structure changes.	Expected to have financial policies (including risk and liquidity management) that create elevated risk of debt restructuring in varied economic environments.	Expected to have financial policies (including risk and liquidity management) that create elevated risk of debt restructuring even in healthy economic environments.

[1] For the linear scoring scale, the Aaa endpoint value is $150 billion. A value of $150 billion or better equates to a numeric score of 0.5. The Ca endpoint value is $0.05 billion. A value of $0.05 billion or worse equates to a numeric score of 20.5.
[2] For the linear scoring scale, the Aaa endpoint value is 0x. A value of 0x equates to a numeric score of 0.5. The Ca endpoint value is 12x. A value of 12x or worse equates to a numeric score of 20.5, as does a negative Debt/EBITDA value.
[3] For the linear scoring scale, when net debt is positive, the Aaa endpoint value is 100%. A value of 100% or better equates to a numeric score of 0.5. The Ca endpoint value is 0%. A value of 0% or worse equates to a numeric score of 20.5. When net debt is negative and RCF is positive, the numeric score is 0.5. When net debt is negative and RCF is negative or zero, the numeric score is 20.5.
[4] For the linear scoring scale, the Aaa endpoint value is 20x. A value of 20x or better equates to a numeric score of 0.5. The Ca endpoint value is (0.5)x. A value of (0.5)x or worse equates to a numeric score of 20.5.
Source: Moody's Ratings

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Sector overview

Telecommunications service providers deliver similar services using similar technologies but are differentiated in many ways, including scale, geographic footprint, market demographics, length of operating history, product mix, network architecture, financial policy, and the structure of their organization, ownership and capital. Companies with stronger credit profiles tend to be very large and well-established, with long operating histories and incumbent market positions, selling a diversified and well-balanced mix of services across leading infrastructure technologies and architectures. Companies with stronger credit profiles also tend to have conservative financial policies, good governance, including strong board structures and policies and procedures, and simple organizational structures. Weaker companies are often smaller and newer industry participants, and they may have more limited product diversity or employ older technologies and architectures. They may also be exposed to higher governance risk, aggressive financial policies or complex corporate structures, and they may have concentrated private equity ownership focused on near-term gains or management teams with limited track records.

The industry is characterized by the emergence of new technologies and high capital intensity. Building, upgrading and maintaining network infrastructure generally requires significant ongoing capital investment. Capital expenditure generally averages 15% to 20% of revenue, but can be much higher or lower depending on a company's market position and investment strategy. Capital intensity is generally driven by (i) the geographic density of customer locations (shorter routes to customers being less costly than longer routes); (ii) topology of footprint (flat unobstructed runs are less costly than those through elevations and obstructions); (iii) infrastructure location (e.g., undersea, underground or aerial), (iv) transmission material and technology (longer-range transmission technologies such as fiber require less equipment such as amplifiers, as well as requiring fewer customer calls and truck rolls for repairs); (v) the extent of competitive pressures; (vi) growth strategies; (vii) government support and subsidies; and (viii) the age of the company's infrastructure.

Competition in the sector has intensified due to the convergence of infrastructure, services, and end-markets. The cancellation of subscriptions to linear multi-channel video content typically available over cable or satellite and its replacement with streaming services or other internet-based broadcasting, i.e., cord-cutting, has driven a significant increase in streaming video content, requiring higher broadband capacity and speeds. Video customers have been shifting their TV programming consumption from the use of one service provider for full-bundle content to the use of tailored bundles for accessing content over the internet. Multiple providers deliver these tailored services direct to the consumer using a wide variety of models, including subscription video on demand and advertising supported video on demand. As this transition has advanced, the demand for high-speed data (e.g., broadband services) has grown exponentially. For many operators, these services have very high profitability following the initial investment. Additionally, aggregate growth in subscribers and pricing is strong, driven by the proliferation of smart devices that stream video and data, the rise of video conferencing and virtual collaboration following a shift to remote working, e-learning, remote healthcare, cloud computing and the emerging opportunities of technologies including the internet of things, autonomous driving, and generative AI.

Many telecommunications service providers invest in similar infrastructure and market opportunities, which often require a mix of both wireline and wireless delivery. As a result, operators with mostly wireline infrastructure may seek to increase their ownership or access to wireless technology (e.g., reselling mobile services as a mobile virtual network operator) and wireless operators may seek to increase their ownership or access to wireline infrastructure. While broadband demand has grown quickly, competition has intensified and market shares have shifted as operators execute strategies to improve their market positioning with an optimal combination of pricing, bundling and service.

The telecommunications sector is subject to a high degree of government regulation, which can change rapidly and meaningfully. In some cases, these changes can be transformative and can significantly affect credit profiles. For example, certain operators that were previously government-approved (and in some cases, government-owned) monopolies, now operate as public or private operators in more deregulated markets and are exposed to greater competition. As privatization and regulatory liberalization have lowered barriers to entry for competitors, the market positions and credit profiles of some operators have suffered.

Discussion of the scorecard factors

In this section, we explain our general approach for scoring each scorecard factor or sub-factor, and we describe why they are meaningful as credit indicators.

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Factor: Scale (10% weight)

Why it matters

Size is an important indicator of the overall depth of a company's business and its success in attracting a variety of customers, as well as its resilience to shocks. Size typically drives, among other aspects, the breadth of a company's customer base, the depth of its business, economies of scale, operational and financial flexibility, and pricing power. Larger companies may have a greater ability to harness business trends, support a stable or growing market position and withstand competitive pressures. Scale can enhance a telecommunications service provider's ability to bundle products and may be accompanied by market leadership, which can bring superior access to customers and positively influence its long-term business viability.

Scale also typically enhances a company's ability to absorb temporary disruptions, acquisitions or misjudgment in the execution of capital investments. Larger companies are generally more broadly diversified, which can help reduce volatility and provide flexibility to generate cash from the divestiture of certain operations, if needed. Larger companies also benefit from greater financial resources as well as access to capital markets, which enhances financial flexibility. These attributes are particularly meaningful in this capital-intensive sector characterized by reduced asset life cycles due to fast-moving technological trends.

How we assess it for the scorecard

REVENUE

Scale is measured (or estimated in the case of forward-looking expectations) using total reported revenue in billions of US dollars.

Factor: Business Profile (25% weight)

Why it matters

A telecommunications service provider's business profile is important because it greatly influences its ability to generate operating cash flows as well as the stability and sustainability of those cash flows. Core aspects of a business profile that drive success or failure typically include the depth and breadth of the company's product offering; the quality and diversity of its customer base; and its competitive landscape, regulatory environment, infrastructure and technology, and market share.

This factor has two sub-factors:

Competitive Position

A company's competitive position is an important differentiator in assessing its long-term sustainability, in particular in the telecommunications sector where substantially different business models coexist. Pricing strategies and power, cost structures, operating margins and capital intensity can vary widely.

Breadth of services

Companies with a broader service offering can generally compete more effectively than can less-diversified operators, which typically receive lower scores for this sub-factor. A diversified service offering typically results in a more robust platform for developing and deploying a range of new products, and for satisfying customer demand, as consumption needs and patterns evolve. Strategic investments in emerging technologies can widen the opportunity for success.

Customer base and geographic diversification

Customer segmentation and geographic reach are also important for an operator's competitive position. A provider with a broader customer base and a larger geographic footprint will typically be more resilient to falling demand or prices in a particular product or market. Serving a broad customer base can also help mitigate rapidly evolving and unfavorable industry trends. Similarly, geographic diversification may help shield new market participants from market disruptions due to unfavorable changes in competitive dynamics, demographics, taxes, regulations, or climate. Changes in regulation or a very uncertain regulatory environment can affect the stability and predictability of the market and can have a significant impact on an operator's success.

Competitive environment

The competitive environment is a meaningful indicator of its resilience to competition. Some telecommunications service providers are long established and hold a very high share of their domestic market. In some jurisdictions, these positions are protected by regulation

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and very high barriers to entry. However, evolving regulatory and legal frameworks in many countries have allowed for and encouraged increased competition.

A company that is the sole or leading operator in its main market, and which is exposed to low competition from alternative providers, is generally in a better position to sustain its profitability and typically has more resources to invest in assets, provide high-quality customer service, and maintain or strengthen its competitive position.

Regulatory environment

Government regulation and oversight is also important in assessing an operator's competitive position. The regulatory framework under which a telecommunications service provider operates influences its ability to earn predictable returns on investments. The essential nature of the service, particularly broadband access, which most governments believe is an essential service, often results in regulatory oversight. In particular, regulators focus on ensuring reliable coverage and high speeds are available, and that the public's expectations for fair competition and competitive pricing are met. There is also a growing dependence on telecommunications service providers' network infrastructure, which has become critical for a wide range of communications in daily activities in society. The unavailability of networks, even for short periods, can adversely affect an individual enterprise and can also severely affect systems in the capital and financial markets, military operations, energy grids and other critical functions.

The balance that regulators in certain regions strike between facilitating competition and protecting employment and price levels in former monopolies can have a major effect on business profiles. The potential for the award of new concessions or licenses and the way regulation enables prospective carriers to build new networks, access other carriers' networks, interconnect their networks with incumbents, and obtain equitable access pricing can heavily influence the competitive landscape and operators' profit margins. For example, in most mature markets incumbents are required to offer access to their infrastructure to new entrants, but they do not have reciprocal access to their competitors' services at regulated prices. Telecommunications service providers are often responsible for making investments in high-speed networks but are unlikely to have exclusive rights to exploit that network, while some new entrants have asset-light businesses and hold dominant positions in their services.

For wireless operators, regulatory decisions to award additional spectrum licenses can increase competition and provide the foundation for the introduction of competing technologies. For services that are price regulated, the sufficiency and predictability of tariffs have a major influence on cash flows, as regulators seek to balance the stabilization or lowering of prices for consumers with companies' ability to earn an adequate return on their network investments.

Infrastructure and technology

The nature and quality of an operator's network infrastructure, including technologies and architectures deployed, are also critical in assessing competitive position. Strong network quality can limit customer churn and create opportunities for market share expansion, while the acquisition and retention of customers is more challenging for operators with weaker networks. For example, we typically view the ownership of, or access to, high-speed technologies such as fiber, which has lower maintenance costs and higher reliability, as a competitive advantage. A mix of delivery methods (e.g., wireless and wireline), wide geographic coverage, and a range of radio frequencies are also crucial to handling increasing traffic volume and reaching a large number of markets.

The wider the spectrum, the more traffic can be carried between mobile sites (the base stations) and mobile phone users. In a specific geographical area, the spectrum is allocated between simultaneous users of the network. Low frequencies allow the signal to be distributed over long distance and to penetrate buildings. These frequencies are typically well suited to the rollout of broad network coverage at relatively low cost. Conversely, higher frequencies are better suited to delivering the capacity required to meet demand for high data rates from a large number of users in urban areas, airports and other densely populated or visited locations.

While the cost of expanding footprints, upgrading older infrastructure and adopting new technologies can be significant, failure to invest at the same pace as competitors can lead to an erosion of market share and revenue.

Market Share

Market share is important because it can provide indications of competitive success, the strength of customer relationships, the potential to benefit from operating leverage and prospects for future performance. The relative positioning of a telecommunications

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service provider within its market segments may provide indications of the sustainability of its operating position and whether it will be able to lead or will be reactive to developments in the industry.

Furthermore, the strength of a telecommunications service provider within its markets can influence customer perceptions, its ability to leverage existing capabilities to develop and support revenue, flexibility to innovate without having to take large risks, and its degree of influence with regulators. In many cases, the large market share of incumbent service providers, combined with established infrastructure and network coverage, has been a significant advantage.

How we assess it for the scorecard

Scoring for the factor is based on two sub-factors: Competitive Position; and Market Share.

COMPETITIVE POSITION

The scoring of this sub-factor is based on a qualitative assessment of the breadth of the company's product offering, the breadth and geographic diversity of its customer base, the extent of competition, the regulatory landscape, and the status of the company's infrastructure and technology.

Breadth of services

The key metric considered for assessing the breadth of a company's services is its revenue mix, i.e., wireless and wireline, voice, data and video (if applicable). We also consider the extent to which the company's suite of services complement each other in delivering customer demand for greater network capacity, higher data transfer speeds, security features that provide resilience against cyberattacks, and cloud storage solutions.

Customer base and geographic diversification

The key metrics considered for assessing a company's customer base and geographic diversification include business and residential customer mix, and the company's footprint internationally, nationally and regionally (i.e., within a very large country such as the US). A telecommunications service provider with a very broad base of business customers and residential customers in different socioeconomic groups, and with a broad footprint across several countries or across several regional markets within a very large country typically receive higher scores than do providers whose customer base is concentrated in a single customer segment, e.g., those with price-sensitive, low-cost customers, or whose geographic area is confined to a one small country.

While we typically consider a company with greater geographic diversity to be more resilient, a company's move into new geographic markets, especially in emerging markets, can expose it to new risks that may weigh on this sub-factor score. These risks can include political or regulatory interference, trapped cash, currency conversion, corruption, or a diversion of management attention away from the company's core strategies and markets.

Competitive environment

In assessing the level of competitive challenge we typically consider revenue trends, the number of operators, rate of subscription change to a particular service relative to demand growth and, for wireless carriers, gross additions, churn (i.e., the percentage of subscribers to a service who discontinue their subscriptions within a given period), and average revenue-per-user trends in the company's core markets.

For multinational operators, we consider the competitive dynamics of each of its major markets. However, our assessment generally focuses on the operator's largest market or markets, as these are often the primary drivers of the company's overall competitive position.

Regulatory environment

In our qualitative assessment of the regulatory environment, we typically consider four characteristics: (i) support for return on investment; (ii) regulatory barriers to entry, such as the propensity for additional licenses or concessions to be issued; (iii) predictability of the regulatory regime; and (iv) the level of reliance on a regulated revenue stream or service subsidies. If a company relies on regulated revenues that might be at risk because of possible changes in regulation, our perception of business risk increases.

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Our assessment of the regulatory framework typically includes consideration of the strength of its political and legal underpinnings, the regulator's track record of predictable and stable decision-making, its independence from political interference, and our forward-looking view on whether these conditions will persist. We typically base our assessment on the relative degree of credit support or challenge a regulatory environment may create for telecommunications service providers in a given jurisdiction, as well as on how this environment or any change to it may affect a specific issuer.

For example, we may consider how the regulatory environment will affect the industry's convergence, and whether regulations tend to favor incumbents or new entrants. We also generally consider a high reliance on a government-regulated revenue stream negatively. The predictability of the regulatory environment is a key aspect for gauging its credit impact. Regulatory uncertainty generally weighs on all companies in a given jurisdiction.

For multinational operators, we typically assess the regulatory frameworks of each country the company operates in. We primarily focus on the largest markets, as regulatory changes in these markets can have a significant impact on an operator's operations.

Infrastructure and technology

We consider the company's exposure to technological advancement and how well-positioned it may be in handling such developments. Our assessment of a company's technology typically includes an evaluation of the lifetime service capabilities and scalability of the company's existing network architecture. In assessing the risk associated with a company's ongoing infrastructure plans, we generally consider the technologies that the company is deploying, especially whether the technologies are proven, the time it will take to deliver the technologies to market, the size of the investment required, and the expected lifetimes of the technologies. While this assessment is largely qualitative, we may consider the capex-to-revenue ratio, with a higher level typically indicating a lower risk of technology obsolescence. A single-digit capex-to-revenue percentage may indicate higher risk.

Generally, we do not expect a given telecommunications service provider's competitive position to exactly match each of the attributes listed for a given scoring category. The sub-factor score is typically the alpha category for which the company has the greatest number of characteristics. However, there may be cases in which one characteristic is sufficiently important to a particular company's credit profile that it has a large influence on the sub-factor score. For example, the benefits of a stable and predictable regulatory environment may be very influential in our scoring even if geographic diversity is limited or the provider does not provide a broad suite of services.

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MARKET SHARE

The scoring of this sub-factor is based on an assessment of the relative market shares a company exhibits in its markets and segments. We use market share information reported by operators or provided directly to us, supplemented by third-party information.

Factor: Profitability and Efficiency (10% weight)

Why it matters

Profits matter because they are needed to generate sustainable cash flow and to maintain a competitive position, including sufficient reinvestment in marketing, research, facilities and human capital. The breadth of business models and diversity of operating environments in the telecommunications service providers sector (e.g., diversified, wireline, wireless, pay TV, regional, national, postpaid or prepaid) makes a multidimensional approach important in assessing profitability. While the level and stability of operating margins is a key consideration in assessing credit quality, revenue trends may also be a driver of a provider's capacity to sustain its profit margins over the medium to long-term. As revenue declines, a company may be able to cut costs to maintain margins on a short-term basis but this may not be achievable indefinitely. Conversely, high margins may be supported by strong revenue growth, as can be the case in some emerging markets, but an operator may have limited pricing power or cost control to mitigate the impact that a slowdown in market dynamics may have on its profit margins. Therefore, the strength of an entity's profitability is generally a function of both the sustainability of its margins and its revenue trajectory.

How we assess it for the scorecard

REVENUE AND MARGIN SUSTAINABILITY

The scoring of this sub-factor is based on a forward-looking qualitative assessment of the trajectory and sustainability of revenue growth and the strength of EBITDA margins. We typically consider the drivers of revenue growth (i.e., from market or organic growth,

or from acquisitions) and the risks inherent in those drivers, and the composition or quality of the margin. For example, we may consider the degree of operational flexibility and capacity and willingness to take the necessary steps to maintain or support margin levels.

Factor: Leverage and Coverage (40% weight)

Why it matters

Leverage and cash flow measures provide important indications of how much financial risk company is willing to undertake. These metrics are also important indicators of a company's ability to sustain its competitive position, invest in growth opportunities and service debt.

This factor comprises three quantitative sub-factors.

Debt / EBITDA

The ratio of total debt to earnings before interest, taxes, depreciation and amortization (Debt/EBITDA) is an indicator of debt serviceability and financial leverage. The ratio is commonly used in this sector as a proxy for comparative financial strength.

RCF / Net Debt

The ratio of retained cash flow to debt (RCF/Net Debt) is an indicator of a company's cash generation (before working capital movements and capital expenditures, and after dividend payments) relative to its net debt (total debt minus cash and cash equivalents).

(EBITDA-Capex) / Interest Expense

The ratio of EBITDA minus capital expenditures to interest expense ((EBITDA-Capex)/Interest Expense) is an indicator of a company's ability to meet its interest obligations and invest in fixed assets.

How we assess it for the scorecard

Scoring for this factor is based on three sub-factors: Debt/EBITDA; RCF/Net Debt; and (EBITDA-Capex)/Interest Expense.

DEBT / EBITDA

The numerator is total debt, and the denominator is EBITDA.

We consider spectrum license payments that are deferred and are made over a number of years in line with their terms as debt amounts. Where such obligations are not reported as debt in the financial statements, we typically adjust debt (as a nonstandard adjustment) to include the outstanding liability.[1]

RCF / NET DEBT

The numerator is retained cash flow (RCF), and the denominator is net debt (total debt minus cash and cash equivalents).

(EBITDA-CAPEX) / INTEREST EXPENSE

The numerator is EBITDA minus capital expenditures, and the denominator is interest expense.

Where spectrum license payments are reported as capex, to enhance peer comparability we typically adjust capex (as a nonstandard adjustment) to exclude these payments.[2]

Factor: Financial Policy (15% weight)

Why it matters

Financial policy encompasses management and board tolerance for financial risk and commitment to a strong credit profile. It is an important rating determinant, because it directly affects debt levels, credit quality, the future direction for the company and the risk of adverse changes in financing and capital structure.

Financial risk tolerance serves as a guidepost to investment and capital allocation. An expectation that management will be committed to sustaining an improved credit profile is often necessary to support an upgrade. For example, we may not upgrade the ratings of

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a company that has built flexibility within its rating category if we believe the company will use that flexibility to fund a strategic acquisition, cash distribution to shareholders, spinoff or other leveraging transaction. Conversely, a company's credit rating may be better able to withstand a moderate leveraging event if management places a high priority on returning credit metrics to pre-transaction levels and has consistently demonstrated the commitment to do so through prior actions. Liquidity management is distinct from the level of liquidity, which is discussed in the "Other considerations" section. Liquidity management is an important aspect of overall risk management and can provide insight into risk tolerance.

Financial policies are very important in the telecommunications service providers sector, which has been characterized by frequent M&A activity. While market consolidation can allow companies to extract value from scale benefits, achieve synergies, improve margins and increase cash flows, debt-financed acquisition activity may heighten credit risk, especially when the acquisition increases the company's business risk profile or distracts management from the core businesses.

Shareholder pressure is also prevalent in some segments of the sector, because established providers often have the capacity to generate significant cash flow, even in the face of declining access lines and challenges in revenue growth. Shareholder activism that results in a large percentage of available cash being used for shareholder distributions such as dividends, equity recapitalizations, or buybacks can weaken a company's credit profile in an environment characterized by increasing competitive challenges and high capital intensity.

How we assess it for the scorecard

We assess the issuer's desired capital structure or targeted credit profile, its history of prior actions, including its track record of risk and liquidity management, and its adherence to its commitments. Attention is paid to management's operating performance, and use of cash flow through different phases of economic and industry cycles. Also of interest is the way in which management responds to key events, such as changes in the credit markets and liquidity environment, legal actions, competitive challenges or regulatory pressures. Considerations include a company's public commitments in this area, its track record of adhering to commitments and our views on the ability of the company to achieve its targets.

When considering event risks in the context of scoring financial policy, we assess the likelihood and potential negative impact of M&A or other types of balance-sheet-transforming events. Management's appetite for M&A activity is assessed, with a focus on the type of transactions (i.e., core competency or new business) and funding decisions. Frequency and materiality of acquisitions and previous financing choices are evaluated. A history of debt-financed or credit-transforming acquisitions generally results in a lower score for this factor. We may also consider negative repercussions caused by shareholders' willingness to sell the company.

We also consider a company's and its owners' past record of balancing shareholder returns and debtholders' interests. A track record of favoring shareholder returns at the expense of debtholders is likely to be viewed negatively in scoring this factor.

Other considerations

Ratings may reflect consideration of additional factors that are not in the scorecard, usually because the factor's credit importance varies widely among the issuers in the sector or because the factor may be important only under certain circumstances or for a subset of issuers. Such factors can include financial controls and the quality of financial reporting; corporate legal structure; the quality and experience of management; assessments of corporate governance as well as environmental and social considerations; exposure to uncertain licensing regimes; and possible government interference in some countries. Regulatory, litigation, liquidity, technology and reputational risk, as well as changes to consumer and business spending patterns, competitor strategies and macroeconomic trends, can also affect ratings.

The following are examples of additional considerations that may be reflected in our ratings and that may cause ratings to be different from scorecard-indicated outcomes.

Environmental, social and governance considerations

Where environmental, social and governance (ESG) issues are meaningful for credit profiles, we incorporate them into our ratings analysis in a variety of ways in the application of our sector-specific methodologies. As one part of our overall credit analysis, we consider how ESG considerations could affect the qualitative and quantitative factors and sub-factors in the scorecard.

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Even where ESG considerations do not affect the measures in the scorecard, or where they cannot be quantified, we incorporate them into our overall analysis of credit drivers that are meaningful to the rating. As a result, we may incorporate these ESG risks qualitatively outside of the scorecard. As part of our ratings analysis, we may establish Issuer Profile Scores (IPSs), which indicate our opinion of the extent to which a given issuer is exposed to E, S and G risks (incorporating ESG-specific mitigants) or benefits from its exposure to E, S or G. The IPSs are inputs to credit ratings. For more information see our methodology that describes our general principles for assessing ESG risks.[3]

Where material, telecommunications service providers' exposure to carbon transition risk and physical climate risk is incorporated in our assessment of scorecard factors and sub-factors, including the Competitive Position and Revenue and Margin Sustainability sub-factors, and in our forward view of financial metrics. The need for more energy-efficient networks may require significant investment in mobile and fixed-line communications network transformation and optimization, which may affect leverage and coverage ratios. Such investments may include software development for server virtualization, smart management, cooling and power storage optimization aimed at reducing power consumption and upgrading of broadband networks from copper cable to optic fiber and upgrading wireless networks to 5G. Issuers may also employ renewable energy to mitigate carbon transition risk. Some issuers in this sector operate in countries with greater exposure to physical climate risks such as hurricanes, typhoons or flooding, which can result in damage to assets or disruption of services.

Some telecommunications service providers face exposure to well-entrenched labor unions with rigid employee regulations, particularly in Europe. Issuers in this sector also face the challenge of monetizing strong demand because consumers have become sensitive to prices in a highly competitive environment. Data security and data privacy issues are prominent social risks in the sector. Telecommunications service providers exchange large amounts of data. A data breach could result in significant legal, regulatory or reputational risk. In addition, issuers in this sector are exposed to increased operational costs to mitigate cyberattacks and to protect private data.

Unlike environmental and social considerations, which have a sectorwide impact, governance considerations are generally issuer specific. Among the areas of focus in corporate governance are audit committee financial expertise, the incentives created by executive compensation packages, related party transactions, interactions with outside auditors, and ownership structure.

Liquidity

Liquidity is an important rating consideration for all telecommunications service providers, and extremely weak liquidity can heavily affect ratings in many cases. However, the relative strength or weakness of liquidity may not have a substantial impact in discriminating between two issuers with an otherwise similar credit profile. Liquidity can be particularly important where companies have large short-term demands on liquidity, for example in highly seasonal operating environments where working capital needs must be considered. We form an opinion on likely near-term liquidity requirements from the perspective of both sources and uses of cash. For more information on our approach, see our liquidity cross-sector methodology.[4]

Management strategy

The quality of management is an important factor supporting a company's credit strength. Assessing the execution of business plans over time can be helpful in assessing management's business strategies, policies, and philosophies and in evaluating management performance relative to performance of competitors and our projections. Management's track record of adhering to stated plans, commitments and guidelines provides insight into management's likely future performance, including in stressed situations.

Asset ownership strategy

While many telecommunications service providers own many of their key infrastructure assets, some providers have partially or fully sold some of their infrastructure assets, including fixed-line networks, data centers and wireless towers, and leased them back. Disposal proceeds from asset sales can allow companies to fund investment in new or expanded infrastructure without needing to raise debt or equity funding, which can be particularly beneficial in times of financial stress. While sales of non-core assets such as passive infrastructure can be credit positive, especially if the proceeds are used to reduce debt, they limit future ability to raise additional cash from disposals. Therefore, companies that continue to own their infrastructure assets generally have greater financial flexibility than those that have sold some or all of their assets. Asset sales can also expose operators to risks related to financing and accounting complexity and potential conflicts of interest.

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Companies can dispose of assets through joint venture arrangements, the sale of minority stakes (see the "Non-wholly owned subsidiaries" section) or on- or off-balance sheet project finance vehicles, e.g., for fiber-optic network rollouts. These transactions can reduce recourse debt but can also result in increased group structure complexity, a credit negative. Financial information related to joint ventures, minority stakes and off-balance-sheet funding vehicles is not always available, which reduces transparency into the company's financial condition.

Financial controls

We rely on the accuracy of audited financial statements to assign and monitor ratings in this sector. The quality of financial statements may be influenced by internal controls, including the proper tone at the top, centralized operations, and consistency in accounting policies and procedures. Auditors' reports on the effectiveness of internal controls, auditors' comments in financial reports and unusual restatements of financial statements or delays in regulatory filings may indicate weaknesses in internal controls.

Excess cash balances

Some companies in this sector may maintain cash balances (meaning liquid short-term investments, as well as cash) that are far in excess of their operating needs. This excess cash, together with the company's policy and motivations, can be an important credit consideration.

The level of cash required to run a business can vary materially based on the region(s) of operation and the specific sub-sectors in which the issuer operates. As a result, cash on the balance sheet is most often considered qualitatively, by assessing the company's track record and financial and liquidity policies. Cash holdings may be a material mitigant in our scenario analyses of potential acquisitions, share buybacks or special dividends. We analyze possible uses of cash, especially for companies that may be under shareholder pressure to return it. By contrast, some companies maintain large cash holdings for long periods in excess of their operating and liquidity needs solely because of conservative financial policies, which provides a stronger indication of an enduring approach that will benefit creditors. Large cash holdings provide some beneficial cushion against credit deterioration, and cash balances are often considered in our analysis of near-term liquidity sources and uses. For more information, see the "Liquidity" section.

While the scorecard in this methodology uses certain leverage and coverage ratios with total (or gross) debt, in cases where we are sufficiently confident that cash will remain on the balance sheet or will be used for creditor-friendly purposes, we may consider excess cash in a more quantitative manner. We do not typically adjust the balance-sheet debt for any specific amount because this implies greater precision than we think is achievable for the uncertain future uses of cash. We may consider creditor-friendly use of cash in our forward view of metrics, for instance, when we expect the cash to be used for debt repayment. We may also cite rating threshold levels for certain companies based on net debt ratios, particularly when these issuers have publicly stated financial targets based on net debt metrics. In cases where we believe that cash on the balance sheet does not confer meaningful credit support, we are more likely to cite gross debt ratios in our credit analysis and rating threshold levels.

Non-wholly owned subsidiaries

Some telecommunications service providers choose to dilute their equity stake in certain material subsidiaries, for example through an initial public offering, which may in some cases negatively impact future financial flexibility. While improving cash holdings on a one-off basis, selling minority interests in subsidiaries may have a negative impact on cash flows available to the parent company that may not be fully reflected in consolidated financial statements. For example, in the case of an equity stake reduction in a subsidiary down to 75%, in the parent's financial statements, all revenue and EBITDA of the subsidiary would typically still be consolidated at the group level.

The parent's share of dividend flows from a non-wholly owned subsidiary is reduced, and minority stakes can increase structural subordination, since dividend flows to minority interest holders are made before the cash flows are available to service debt at the parent company. While less frequent, sale of a minority stake may be accompanied by policies protective of the subsidiary that further limit the parent's financial flexibility, for instance restrictions on cash pooling with other members of the corporate family, limitations on dividends and distributions, or arms-length business requirements. Minority stakeholders may have seats on the board of the subsidiary.

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In many cases, we consider the impact of non-wholly owned subsidiaries qualitatively. However, in some cases we may find that an additional view of financial results, such as analyzing cash flows on a proportional consolidation basis, may be very useful to augment our analysis based on consolidated financial statements. When equity dilution or structural subordination arising from non-wholly owned subsidiaries is material and negative, the credit impact is captured in ratings but may not be fully reflected in scorecard-indicated outcomes.

For companies that hold material minority interest stakes, consolidated funds from operations typically includes the dividends received from the minority subsidiary, while none of its debt is consolidated. When such dividends are material to the company's cash flows, these cash flows may be subject to interruption if they are required for the minority subsidiary's debt service, capital expenditures or other cash needs. When minority interest dividends are material, we may also find that proportional consolidation or another additional view of financial results is useful to augment our analysis of consolidated financials. We would generally also consider structural subordination in these cases. Proportional consolidation brings a portion of the minority subsidiary's debt onto the balance sheet, but this debt is structurally senior to debt at the parent company, because it is closer to the assets and cash flows of the minority subsidiary. When these credit considerations are material, their impact is captured in ratings but may not be fully reflected in scorecard-indicated outcomes.

Event risk

We also recognize the possibility that an unexpected event could cause a sudden and sharp decline in an issuer's fundamental creditworthiness, which may cause actual ratings to be lower than the scorecard-indicated outcome. Event risks — which are varied and can range from leveraged recapitalizations to sudden regulatory changes or liabilities from an accident — can overwhelm even a stable, well-capitalized firm. Some other types of event risks include M&A, asset sales, spinoffs, shareholder distributions, litigation, pandemics, significant cybercrime events and geopolitical conflicts.

Additional metrics

The metrics included in the scorecard are those that are generally most important in assigning ratings to companies in this sector; however, we may use additional metrics to inform our analysis of specific companies. These additional metrics may be important to our forward view of metrics that are in the scorecard or other rating factors.

For example, free cash flow is not always an important differentiator of credit profiles. Strong companies with excellent investment opportunities may demonstrate multiyear periods of negative free cash flow while retaining solid access to capital and credit, because these investments will yield stable cash flows in future years. Weaker companies with limited access to credit may have positive free cash flow for a period of time because they have curtailed the investments necessary to maintain their assets and future cash-generating prospects. However, in some cases, free cash flow can be an important driver of the future liquidity profile of an issuer, which, as noted above, can have a meaningful impact on ratings.

Parental support

Ownership can provide ratings lift for a particular company in the telecommunications service providers sector if it is owned by a highly rated owner(s) and is viewed to be of strategic importance to those owners. In our analysis of parental support, we typically consider whether the parent has the financial capacity and strategic incentives to provide support to the issuer in times of stress or financial need (e.g., a major capital investment or advantaged operating agreement), or has already done so in the past. Conversely, if the parent puts a high dividend burden on the issuer, which in turn reduces its flexibility, the ratings would reflect this risk.

Government-related issuers may receive ratings uplift due to expected government support. However, for certain issuers, government ownership can have a negative impact on the underlying Baseline Credit Assessment.[5] For example, price controls, onerous taxation and high distributions can have a negative effect on an issuer's underlying credit profile.

Other institutional support

In some countries, large corporate issuers have received government or banking support in the event of financial difficulties because of their overall importance to the functioning of the economy. In Japan, our corporate ratings consider the support that has operated there for large and systemically important organizations. Over the years, this has resulted in lower levels of default than might otherwise have occurred. Our approach considers whether the presence of group and banking relationships may provide support when systemically important companies encounter significant financial stress.

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Using the scorecard to arrive at a scorecard-indicated outcome

1. Measurement or estimation of factors in the scorecard

In the "Discussion of the scorecard factors" section, we explain our analytical approach for scoring each scorecard factor or sub-factor, and we describe why they are meaningful as credit indicators. When a factor comprises sub-factors, we score at the sub-factor level. Some factors do not have sub-factors, in which case we score at the factor level.

The information used in assessing the sub-factors is generally found in or calculated from information in the company's financial statements or regulatory filings, derived from other observations or estimated by Moody's analysts. We may also incorporate non-public information.

Our ratings are forward-looking and reflect our expectations for future financial and operating performance. However, historical results are helpful in understanding patterns and trends of a company's performance as well as for peer comparisons. Financial ratios,[6] unless otherwise indicated, are typically calculated based on an annual or 12-month period. However, the factors in the scorecard can be assessed using various time periods. For example, rating committees may find it analytically useful to examine both historical and expected future performance for periods of several years or more.

All of the quantitative credit metrics incorporate our standard adjustments[7] to income statement, cash flow statement and balance sheet amounts for items such as underfunded pension obligations and operating leases. We may also make other analytical adjustments that are specific to a particular company.

2. Mapping scorecard factors to a numeric score

After estimating or calculating each factor or sub-factor, each outcome is mapped to a broad Moody's rating category (Aaa, Aa, A, Baa, Ba, B, Caa or Ca, also called alpha categories) and to a numeric score.

Qualitative factors are scored based on the description by broad rating category in the scorecard. The numeric value of each alpha score is based on the scale below.

Exhibit 3
Numeric equivalents for qualitative factor and sub-factor scores

Aaa	Aa	A	Baa	Ba	B	Caa	Ca
1	3	6	9	12	15	18	20

Source: Moody's Ratings

Quantitative factors are scored on a linear continuum. For each metric, the scorecard shows the range by alpha category. We use the scale below and linear interpolation to convert the metric, based on its placement within the scorecard range, to a numeric score, which may be a fraction. As a purely theoretical example, if there were a ratio of revenue to interest for which the Baa range was 50x to 100x, then the numeric score for an issuer with revenue/interest of 99x, relatively strong within this range, would score closer to 7.5, and an issuer with revenue/interest of 51x, relatively weak within this range, would score closer to 10.5. In the text or table footnotes, we define the endpoints of the line (i.e., the value of the metric that constitutes the lowest possible numeric score, and the value that constitutes the highest possible numeric score).

Exhibit 4
Ranges of numeric equivalents for quantitative factor and sub-factor scores

Aaa	Aa	A	Baa	Ba	B	Caa	Ca
0.5 - 1.5	1.5 - 4.5	4.5 - 7.5	7.5 - 10.5	10.5 - 13.5	13.5 - 16.5	16.5 - 19.5	19.5 - 20.5

Source: Moody's Ratings

3. Determining the overall scorecard-indicated outcome

The numeric score for each sub-factor (or each factor, when the factor has no sub-factors) is multiplied by the weight for that sub-factor (or factor), with the results then summed to produce an aggregate numeric score. The aggregate numeric score is then mapped back to a scorecard-indicated outcome based on the ranges in the table below.

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Exhibit 5
Scorecard-indicated outcome

Aggregate numeric score	Scorecard-indicated outcome
x ≤ 1.5	Aaa
1.5 < x ≤ 2.5	Aa1
2.5 < x ≤ 3.5	Aa2
3.5 < x ≤ 4.5	Aa3
4.5 < x ≤ 5.5	A1
5.5 < x ≤ 6.5	A2
6.5 < x ≤ 7.5	A3
7.5 < x ≤ 8.5	Baa1
8.5 < x ≤ 9.5	Baa2
9.5 < x ≤ 10.5	Baa3
10.5 < x ≤ 11.5	Ba1
11.5 < x ≤ 12.5	Ba2
12.5 < x ≤ 13.5	Ba3
13.5 < x ≤ 14.5	B1
14.5 < x ≤ 15.5	B2
15.5 < x ≤ 16.5	B3
16.5 < x ≤ 17.5	Caa1
17.5 < x ≤ 18.5	Caa2
18.5 < x ≤ 19.5	Caa3
19.5 < x ≤ 20.5	Ca
x > 20.5	C

Source: Moody's Ratings

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For example, an issuer with an aggregate numeric score of 11.7 would have a Ba2 scorecard-indicated outcome.

In general, the scorecard-indicated outcome is oriented to the corporate family rating (CFR) for speculative-grade issuers and to the senior unsecured rating for investment-grade issuers. For issuers that benefit from rating uplift from parental support, government ownership or other institutional support, we consider the underlying credit strength or Baseline Credit Assessment for comparison to the scorecard-indicated outcome. For an explanation of the Baseline Credit Assessment, please refer to *Rating Symbols and Definitions* and to our cross-sector methodology for government-related issuers.[8]

Assigning issuer-level and instrument-level ratings

After considering the scorecard-indicated outcome, other considerations and relevant cross-sector methodologies, we typically assign a CFR to speculative-grade issuers or a senior unsecured rating for investment-grade issuers. For issuers that benefit from rating uplift from government ownership, we may assign a Baseline Credit Assessment.[9]

Individual debt instrument ratings may be notched up or down from the CFR or the senior unsecured rating to reflect our assessment of differences in expected loss related to an instrument's seniority level and collateral. The documents that provide broad guidance for such notching decisions are the rating methodology on loss given default for speculative-grade nonfinancial companies, the methodology for notching corporate instrument ratings based on differences in security and priority of claim, and the methodology for assigning short-term ratings.[10]

Key rating assumptions

For information about key rating assumptions that apply to methodologies generally, please see *Rating Symbols and Definitions*.[11]

Limitations

In the preceding sections, we have discussed the scorecard factors and many of the other considerations that may be important in assigning ratings. In this section, we discuss limitations that pertain to the scorecard and to the overall rating methodology.

Limitations of the scorecard

There are various reasons why scorecard-indicated outcomes may not map closely to actual ratings.

The scorecard in this rating methodology is a relatively simple reference tool that can be used in most cases to approximate credit profiles of companies in this sector and to explain, in summary form, many of the factors that are generally most important in assigning ratings to these companies. Credit loss and recovery considerations, which are typically more important as an issuer gets closer to default, may not be fully captured in the scorecard. The scorecard is also limited by its upper and lower bounds, causing scorecard-indicated outcomes to be less likely to align with ratings for issuers at the upper and lower ends of the rating scale.

The weights for each factor and sub-factor in the scorecard represent an approximation of their importance for rating decisions across the sector, but the actual importance of a particular factor may vary substantially based on an individual company's circumstances.

Factors that are outside of the scorecard, including those discussed above in the "Other considerations" section, may be important for ratings, and their relative importance may also vary from company to company. In addition, certain broad methodological considerations described in one or more cross-sector rating methodologies may be relevant to ratings in this sector.[12] Examples of such considerations include the following: how sovereign credit quality affects non-sovereign issuers, the assessment of credit support from other entities, the relative ranking of different classes of debt and hybrid securities, and the assignment of short-term ratings.

We may use the scorecard over various historical or forward-looking time periods. Furthermore, in our ratings we often incorporate directional views of risks and mitigants in a qualitative way.

General limitations of the methodology

This methodology document does not include an exhaustive description of all factors that we may consider in assigning ratings in this sector. Companies in the sector may face new risks or new combinations of risks, and they may develop new strategies to mitigate risk. We seek to incorporate all material credit considerations in ratings and to take the most forward-looking perspective that visibility into these risks and mitigants permits.

Ratings reflect our expectations for an issuer's future performance; however, as the forward horizon lengthens, uncertainty increases and the utility of precise estimates, as scorecard inputs or in other considerations, typically diminishes. Our forward-looking opinions are based on assumptions that may prove, in hindsight, to have been incorrect. Reasons for this could include unanticipated changes in any of the following: the macroeconomic environment, general financial market conditions, industry competition, disruptive technology, or regulatory and legal actions. In any case, predicting the future is subject to substantial uncertainty.

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Moody's related publications

Credit ratings are primarily determined through the application of sector credit rating methodologies. Certain broad methodological considerations (described in one or more cross-sector rating methodologies) may also be relevant to the determination of credit ratings of issuers and instruments. A list of sector and cross-sector credit rating methodologies can be found here.

For data summarizing the historical robustness and predictive power of credit ratings, please click here.

For further information, please refer to *Rating Symbols and Definitions*, which is available here.

Moody's Basic Definitions for Credit Statistics (User's Guide) can be found here.

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Endnotes

1 For an explanation of our standard adjustments, please see the cross-sector methodology that describes our financial statement adjustments in the analysis of nonfinancial corporations. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

2 For an explanation of our standard adjustments, please see the cross-sector methodology that describes our financial statement adjustments in the analysis of nonfinancial corporations. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

3 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

4 A link to a list of our cross-sector methodologies can be found in the "Moody's related publications" section.

5 For an explanation of the Baseline Credit Assessment, please refer to *Rating Symbols and Definitions* and to our cross-sector methodology for government-related issuers. A link to a list of our sector and cross-sector methodologies and a link to *Rating Symbols and Definitions* can be found in the "Moody's related publications" section.

6 For definitions of our most common ratio terms, please see *Moody's Basic Definitions for Credit Statistics (User's Guide)*. A link can be found in the "Moody's related publications" section.

7 For an explanation of our standard adjustments, please see the cross-sector methodology that describes our financial statement adjustments in the analysis of nonfinancial corporations.

8 A link to a list of our sector and cross-sector methodologies and a link to *Rating Symbols and Definitions* can be found in the "Moody's related publications" section.

9 For an explanation of the Baseline Credit Assessment, please refer to *Rating Symbols and Definitions* and to our cross-sector methodology for government-related issuers. A link to a list of our sector and cross-sector methodologies and a link to *Rating Symbols and Definitions* can be found in the "Moody's related publications" section.

10 A link to a list of our sector and cross-sector rating methodologies can be found in the "Moody's related publications" section.

11 A link to *Rating Symbols and Definitions* can be found in the "Moody's related publications" section.

12 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

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REPORT NUMBER 1375330

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